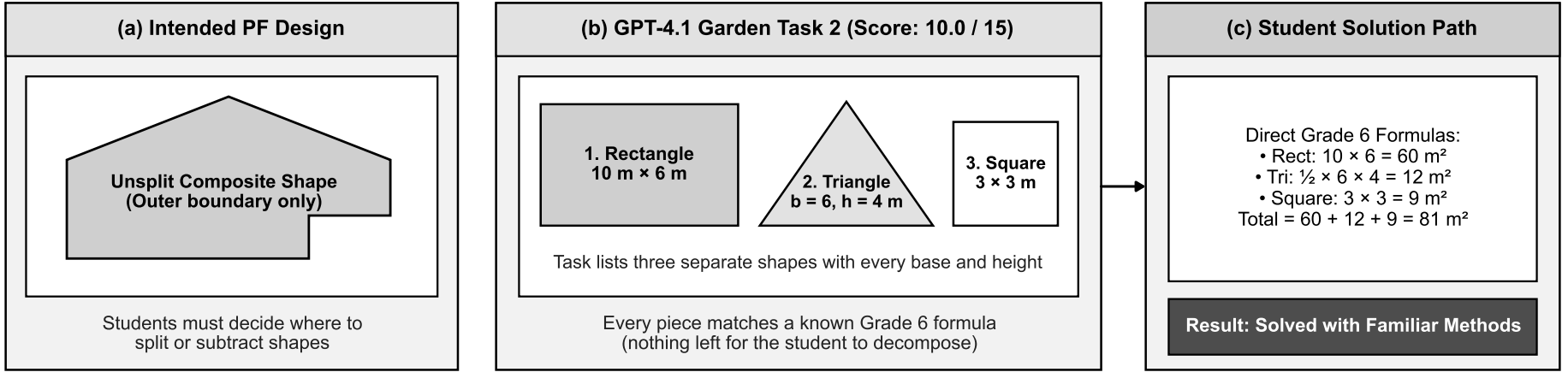
(a) Intended PF Design
Unsplit Composite Shape
(Outer boundary only)
Students must decide where to split or subtract shapes
(b) GPT-4.1 Garden Task 2 (Score: 10.0 / 15)
1. Rectangle
10 m × 6 m
2. Triangle
b = 6, h = 4 m
3. Square
3 × 3 m
Task lists three separate shapes with every base and height
Every piece matches a known Grade 6 formula
(nothing left for the student to decompose)
(c) Student Solution Path
Direct Grade 6 Formulas:
• Rect: 10 × 6 = 60 m²
• Tri: ½ × 6 × 4 = 12 m²
• Square: 3 × 3 = 9 m²
Total = 60 + 12 + 9 = 81 m²
Result: Solved with Familiar Methods